CIS437 - Cloud Computing

– Geographic Issues in Cloud Computing –

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Based on the original version by Professor Erik Fredericks

Last Class: Serverless Functions

- Serverless Computing and FaaS
- How Serverless Functions Work
- Triggers and Common Use Cases
- Benefits, Trade-offs, and Cold Starts
- Function Design and Access Control
- Event-Driven Functions
- Serverless Cost

Agenda

- Regions and Zones
- Choosing Cloud Locations
- Latency and Availability
- Single-Region and Multi-Region Architecture
- Data Residency and Geographic Restrictions
- Geographic Access and Traffic Control
- Load Balancing
- Practice

FIRST, SOME IN-CLASS WORK



Three concerns that may arise from deploying apps globally

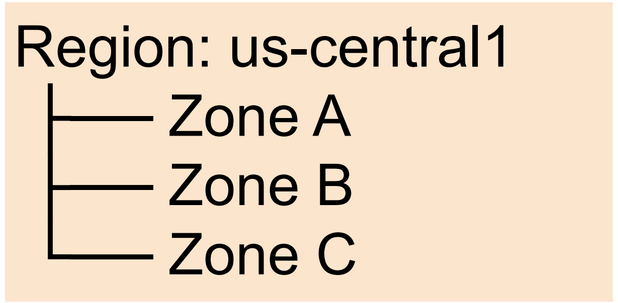
Why Does Geography Matter?

- Latency
- Availability and resilience
- Data residency and compliance
- Service availability
- Cost and data transfer
- Geographic access restrictions

Regions and Zones

Region:

- Independent geographic area containing multiple zones
- Example: us-east1



Zone:

- Deployment area within a region
- A region usually contains multiple zones
- Example: us-east1-a

AWS and Azure use similar region/zone concepts, although terminology and details differ.

Some cloud resources are zonal (e.g. VM); others are regional.

Regions and zones

You Have Already Been Choosing Locations

Every time you create a cloud resource, you often choose **where it is hosted**.

Examples:

- VM → zone / region
- Storage → location / region
- Managed service → supported region



Can we serve users from a closer region?

- Yes, but the architecture must be configured to do that.

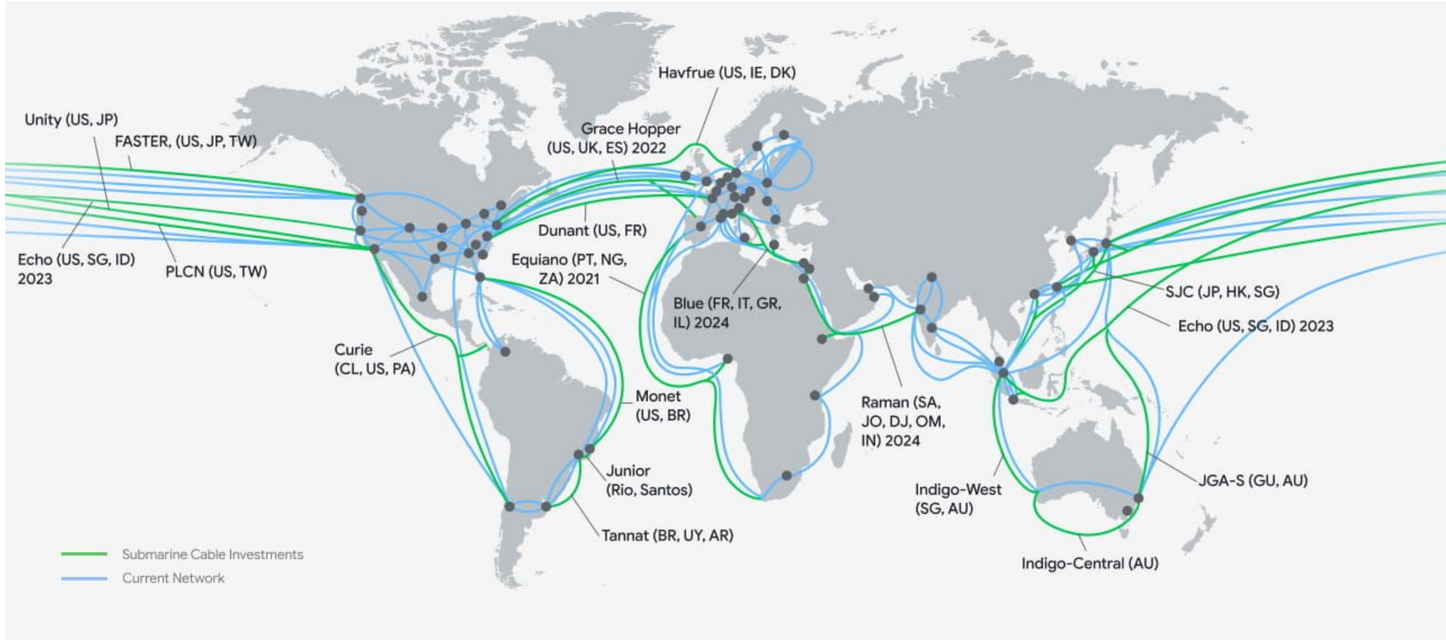
Choosing a Region: What Matters?

- Distance to users → latency
- Availability requirements
- Data residency / compliance
- Service availability in that region
- Cost
- Location of related services and data

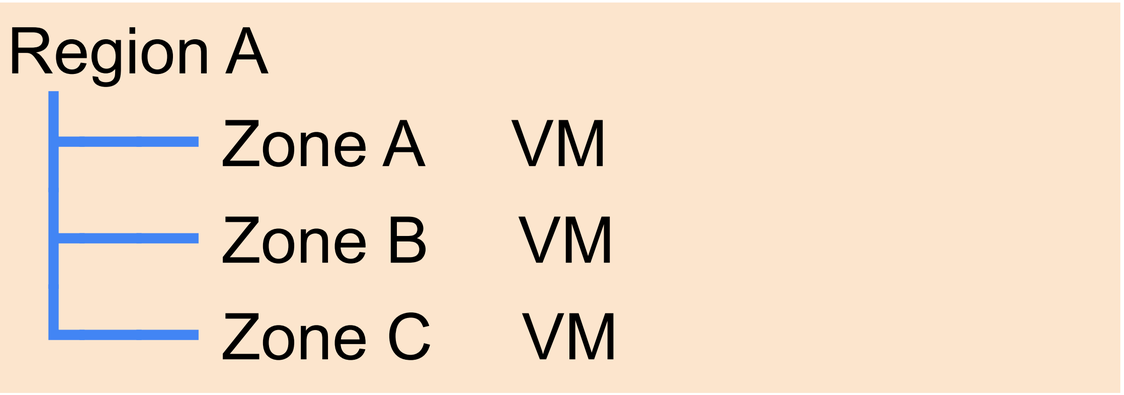
Location-Based Serving: More Than Just Speed

When serving users from different locations, consider:

- Where are the users?
- Where is the application running?
- Where is the data stored?
- Who is allowed to access it?
- Are we optimizing only for latency, or also for availability, compliance, and cost?



Availability: Zone vs. Region Failure



If everything is in one zone

- A zone failure can take down the application

If resources span multiple zones

- Another zone may continue serving users if the application is designed for multi-zone failover

Multiple resources in one zone ≠ zone-level redundancy

What If an Entire Region Fails?



Multiple zones protect against some zone-level failures.

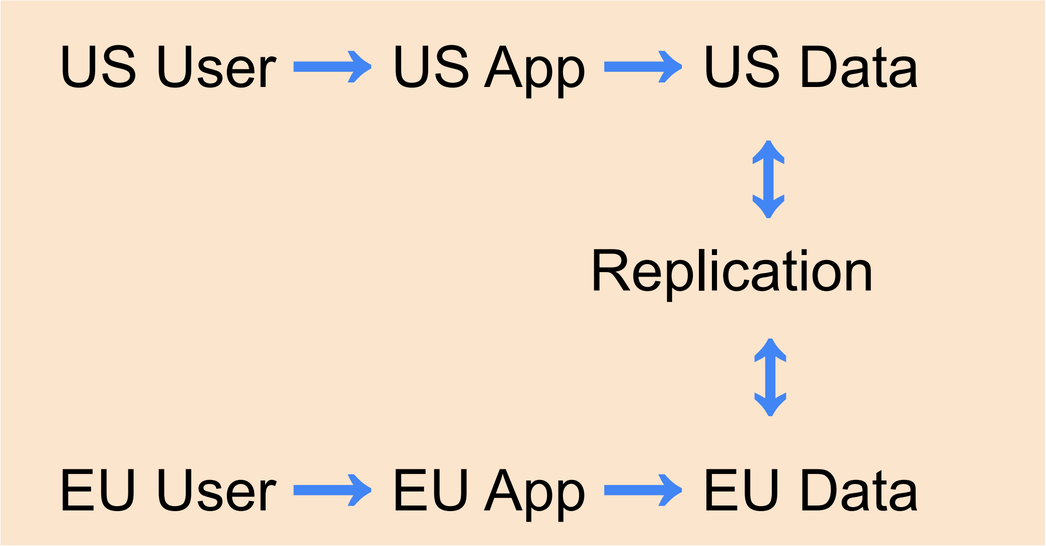
To tolerate a regional outage, the application must be designed to run in another region.

Multi-region resilience requires both application failover and data availability.

Single-Region vs. Multi-Region Architecture

	Single Region	Multi-Region
Architecture	Simpler	More complex
Cost	Usually lower	Usually higher
Region failure	Greater impact	Can fail over to another region if designed for it
Data management	Simpler	More complex; replication/synchronization may be needed
Latency	Good for nearby users	Can reduce latency for geographically distributed users

Multi-Region Data Replication



Replication creates new questions:

- How quickly should updates propagate between regions?
- What happens if two regions update the same data at the same time?
- How much does replication cost?
- Are copies allowed in both locations?

Data Residency / Transfer Policies

Data residency: requirements or policies about **where data is stored and processed geographically**.

- Where could this become an issue?

Microsoft: Data Location

Where Does Data Residency Matter?

Data residency - where the data is stored

- Where would this be an issue?

Private information

- Data we might be concerned about:
  - Medical
  - Financial
  - Military
  - …
  - just plain old private data



Different Locations, Different Rules

Examples:

- Privacy and data-protection requirements, such as GDPR
- Rules for advertising or online services involving children
- Gambling, gaming, and loot-box regulations

Different countries or regions may require you to:

- Restrict access
- Handle data differently
- Serve different content or features

Cloud Providers Give You Location Controls

- Many cloud resources let you select a region or location
- Some services support multi-region configurations
- Providers offer tools for access, encryption, logging, and policy enforcement
- Customers are still responsible for choosing configurations that meet their requirements

"Microsoft provides strong customer commitments regarding cloud services data residency and transfer policies. Most Azure services are deployed regionally and enable the customer to specify the region into which the service will be deployed, e.g., the United States. This commitment helps ensure that Customer Data stored in a U.S. region will remain in the United States and will not be moved to another region outside the United States." —---------—------- Azure: Managing export controls

"Large-scale, multinational CSPs, often called hyperscale CSPs, represent a transformational disruption in technology because of how they support their customers with high degrees of efficiency, agility, and innovation as part of world-class security offerings. The whitepaper explains how hyperscale CSPs, such as AWS, that might be located out of country provide their customers the ability to achieve high levels of data protection through safeguards on their own platform and with turnkey tooling for their customers. They do this while at the same time preserving nation-state regulatory sovereignty." —---------—------- Addressing Data Residency with AWS

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"**Customers** with Export Administration Regulations (EAR)-regulated workloads **are responsible** for determining what steps (if any) are necessary to ensure that their use of the cloud is consistent with the EAR. If customers choose to deploy any of the above technologies to meet their EAR compliance requirements, **customers have final responsibility** to properly deploy and maintain them." ---- Google: EAR

Export Controls and Geographic Restrictions

Some technology, software, and data may be subject to restrictions on:

- Where they can be transferred or accessed
- Who can access them
- How they can be used

Geographic location can therefore affect cloud architecture.

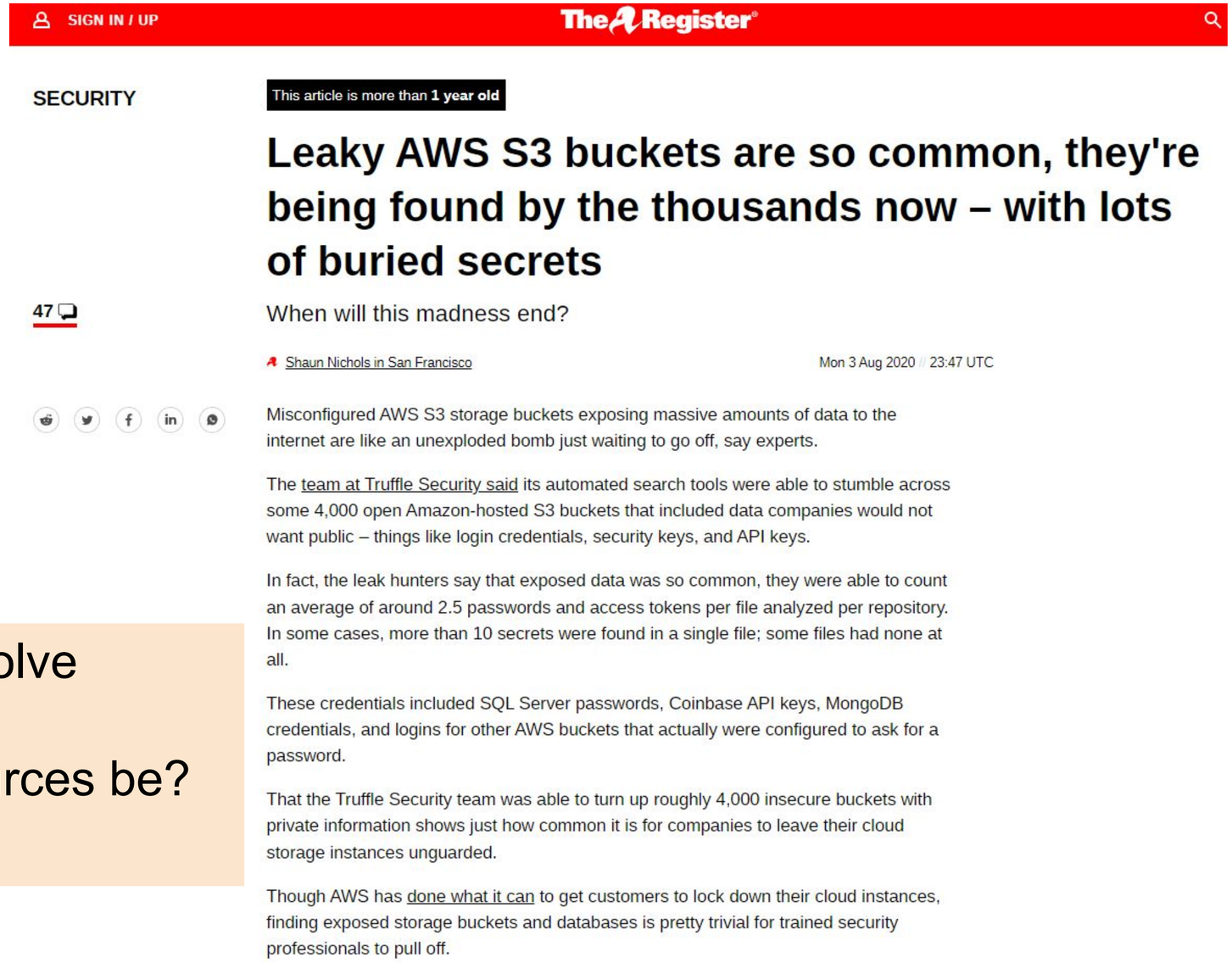
Rules change over time — check current official guidance.

Export Administration Regulations (EAR)

Geography Does Not Replace Security

Choosing the correct region does **not** make data secure.

A storage resource can still be exposed through incorrect permissions.



SIGN IN / UP

The Register

SECURITY

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This article is more than **1 year old**

Leaky AWS S3 buckets are so common, they're being found by the thousands now – with lots of buried secrets

When will this madness end?

Shaun Nichols in San Francisco — Mon 3 Aug 2020 // 23:47 UTC

Misconfigured AWS S3 storage buckets exposing massive amounts of data to the internet are like an unexploded bomb just waiting to go off, say experts.

The team at Truffle Security said its automated search tools were able to stumble across some 4,000 open Amazon-hosted S3 buckets that included data companies would not want public – things like login credentials, security keys, and API keys.

In fact, the leak hunters say that exposed data was so common, they were able to count an average of around 2.5 passwords and access tokens per file analyzed per repository. In some cases, more than 10 secrets were found in a single file; some files had none at all.

These credentials included SQL Server passwords, Coinbase API keys, MongoDB credentials, and logins for other AWS buckets that actually were configured to ask for a password.

That the Truffle Security team was able to turn up roughly 4,000 insecure buckets with private information shows just how common it is for companies to leave their cloud storage instances unguarded.

Though AWS has done what it can to get customers to lock down their cloud instances, finding exposed storage buckets and databases is pretty trivial for trained security professionals to pull off.

Location controls and security controls solve different problems.

- **Geography:** Where can the data/resources be?
- **Security:** Who can access them?

Controlling Access

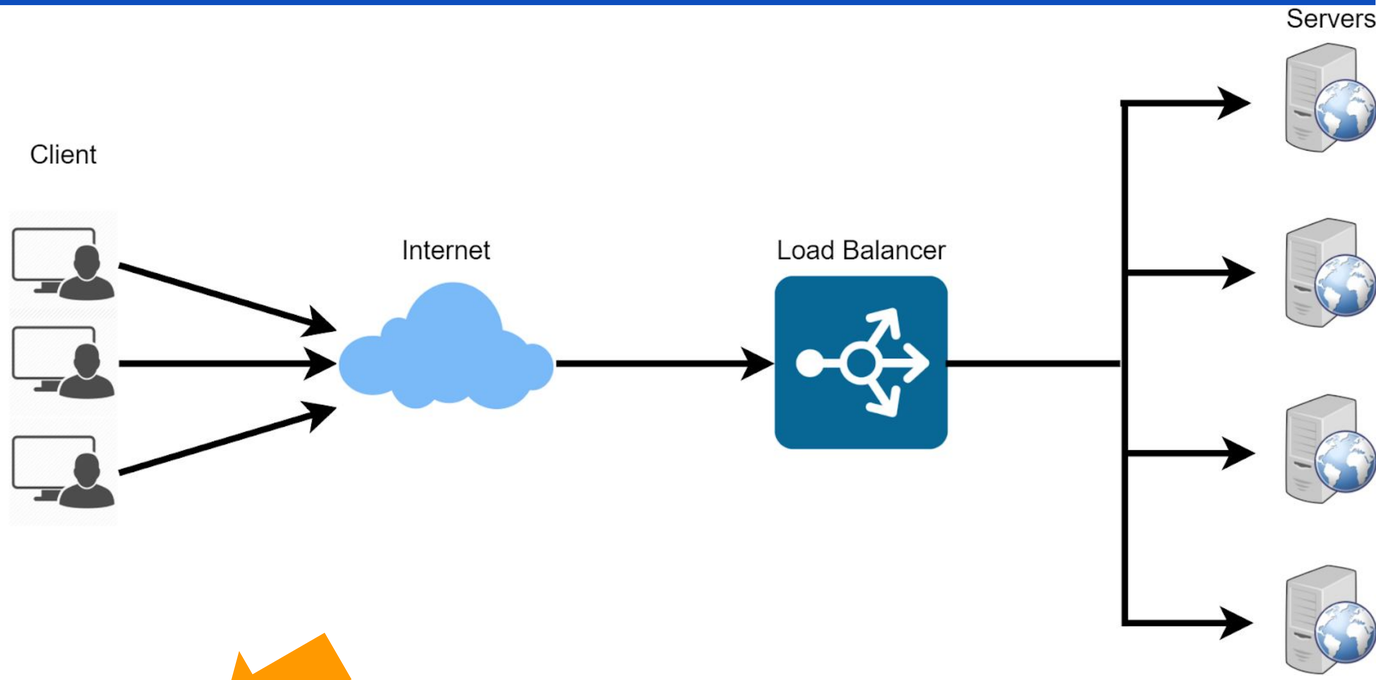
Prevent unnecessary access

- IAM / least privilege
- Limit public access
- Restrict IP ranges when appropriate
- Disable unused services

Detect problems

- Log access
- Monitor unexpected activity

Allow or deny traffic from a specific region



Global Load Balancer

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US Region App

EU Region App

Set up a regional external passthrough Network Load Balancer with a backend service

Lab: Creating Cross-region Load Balancing

In this lab, you will

- Deploy backend resources in different locations
- Configure a load balancer
- Route traffic to the backend resources
- Observe how load balancing supports a geographically distributed application

The lab can be found **here**.



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Summary

- Cloud resources are deployed in geographic locations
- Regions contain multiple zones
- Location affects latency, availability, cost, and service access
- Multi-zone and multi-region designs can improve resilience but add complexity
- Data replication introduces consistency, cost, and residency questions
- Data residency and regulations can restrict where data is stored, processed, or accessed
- IAM, Cloud Armor, and load balancing help implement access and geographic architecture decisions

Practice



Scenario 1: An application has three VMs, but all three are in the same zone. Is it protected against a failure of that entire zone? Why or why not?

Scenario 2: A company must keep EU customer data in approved European locations. What factor should strongly influence its choice of cloud region?

Scenario 3: An application serves users in both the US and Europe, but all application servers are in the US. European users experience high latency. What architectural change could help?

Scenario 4: A company wants to deny requests originating from selected geographic regions. What type of cloud control could help implement this?

Scenario 5: An application runs in two regions and both regions can update customer data. What new data-management problem can arise?